08-23 第2篇

阅读理解（说明文） Why Do Cats Almost Always Land on Their Feet?

① Many people know the old saying: "A cat has nine lives." Of course no cat really has nine lives, but if you have ever seen a young **cat** jump off a sofa, or even fall from a tree, you might believe it for a moment. Somehow the cat **twists** in the air and lands quietly on its feet, as if nothing had happened. This special skill is not magic. It comes from a clever body and a very short, but perfect, sequence of moves — called the **"cat righting reflex"**.

② To understand how a cat does this, we first need to look at two things: its **body shape** and its **inner ear**.

③ A cat's body is **not like ours**. Its backbone is made of many small bones with very flexible joints between them. A human can bend about 180 degrees forward; a cat can **twist its body almost like a wet towel**, rotating the upper half and the lower half in different directions at the same time. This flexibility is the first secret.

④ The second secret is in the cat's **ear**. Deep inside the ear there is a tiny organ called the **vestibular system**, a set of small tubes filled with fluid. When the cat's head turns, the fluid also moves, and tiny hairs inside the tubes feel this movement. The cat's brain then knows, in less than a second, "My head is now upside down." This organ works in humans too, but in cats it is especially sensitive and fast.

⑤ Now, what actually happens when a cat begins to fall? Scientists who filmed falling cats with high-speed cameras noticed a clear pattern:

- **Step 1 — Feel the fall.** The ear tells the brain which way is "up" and which way is "down".
- **Step 2 — Turn the head first.** The cat quickly rotates its head, so that its face points down toward the ground. The eyes check the direction; the head acts like a compass.
- **Step 3 — Twist the upper body.** The front legs and shoulders follow the head and rotate toward the ground.
- **Step 4 — Twist the lower body.** The back legs and tail rotate last, catching up with the front. Some cats use their long tails to help balance, much like a tightrope walker uses a long pole.
- **Step 5 — Prepare the legs.** Just before landing, the cat stretches its legs out and softens them like four small **springs**, ready to absorb the force.

⑥ The whole sequence takes about **half a second**. A cat needs a **minimum height** — usually around 60 centimetres — to have enough time to complete the turn. If the cat falls from a very low place, like the middle of a chair, it may land with its back or side instead, because the body did not have time to rotate fully.

⑦ But this does not mean cats can fall safely from **any** height. A fall from the 10th floor is still very dangerous, and many city cats are hurt this way every year. What is interesting, though, is that from very tall buildings, the cat has time to do something else: it **relaxes** and **stretches all four legs out**, almost like a flying squirrel. This spreads the force over a larger area when the cat finally lands. Some vets even say that cats can be more hurt from a fall of four floors than from ten floors, because four floors give enough time to get scared but not enough time to fully relax.

⑧ Next time you see a cat leap safely from a wall, remember: behind that quiet, elegant moment there is a small but amazing science. Hundreds of nerves in the ear, a backbone like a soft rope, and half a second of perfect timing — all working quietly together, so that a cat can land on its feet and walk away as if nothing had happened.

注释： twist 扭转；reflex 反射；vestibular system 前庭系统；absorb 吸收；vet 兽医

根据文章内容，选择最佳答案。

1. What is the "cat righting reflex" mentioned in the passage?

A. A magical power that gives every cat nine lives.
B. A short, perfect sequence of moves that lets a falling cat turn and land on its feet.
C. A special skill only adult tigers have.
D. A computer program for testing cats.

2. Why is a cat's backbone important for the righting reflex?

A. Because it is made of glass and never breaks.
B. Because it is made of many small bones with flexible joints, so the cat can twist its upper and lower body in different directions at the same time.
C. Because it is stronger than steel.
D. Because it has the same shape as a human spine.

3. What does the vestibular system inside the cat's ear do?

A. It helps the cat hear human voices far away.
B. It keeps the cat's fur clean.
C. It uses fluid and tiny hairs to feel head movement, so the brain quickly knows which way is up or down.
D. It produces electric power for the brain.

4. Why does a cat need a minimum height (around 60 cm) to land safely on its feet?

A. Because it is shy about short falls.
B. Because it needs enough time to complete the full sequence of turning movements before landing.
C. Because short falls have too much wind.
D. Because the floor is harder in low places.

5. Why can a cat sometimes be more hurt by a fall of four floors than by a fall of ten floors, according to some vets?

A. Because four floors have more gravity than ten floors.
B. Because from very tall buildings, the cat has time to fully relax and stretch out its four legs, spreading the force; four floors only give it time to be scared.
C. Because every cat chooses to jump only from buildings taller than ten floors.
D. Because low floors have stronger wind.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第①段末 "the cat righting reflex — a clever body and a very short, but perfect, sequence of moves"
【解析】文中直接给出定义。A 属俗语形象化表述，不是定义；C、D 均无依据。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "many small bones with very flexible joints ... rotating the upper half and the lower half in different directions at the same time"
【解析】脊柱的灵活关节允许上下身同时反向旋转。A、C、D 均与原文不符。故选 B。

3.
【答案】C
【题型】细节理解题
【定位】第④段 "small tubes filled with fluid ... tiny hairs inside the tubes feel this movement. The cat's brain then knows ... My head is now upside down"
【解析】前庭系统靠液体和毛细胞感知头部运动。A、B、D 均与原文不符。故选 C。

4.
【答案】B
【题型】推理判断题
【定位】第⑥段 "takes about half a second. A cat needs a minimum height ... to have

enough time to complete the turn"
【解析】旋转需要时间，因此需要一定的最低高度。A、C、D 均无依据。故选 B。

5.
【答案】B
【题型】细节理解题
【定位】第⑦段 "from very tall buildings ... relaxes and stretches all four legs out ... four floors give enough time to get scared but not enough time to fully relax"
【解析】原文明确说明很高楼放松伸展可分散冲击。A、C、D 均与原文不符。故选 B。